CONSTRUCTOR UNIVERSITY

**MDSSB-DSOC-02 Data Science Concepts**

# Representative Samples and Large-scale Surveys

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# Large-scale surveys

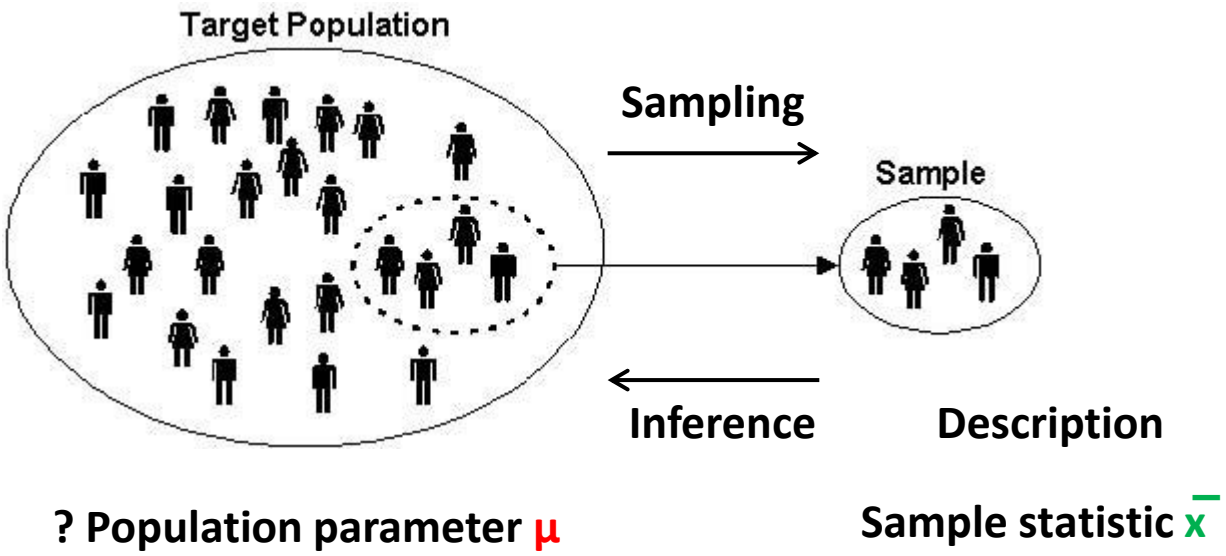
- Survey drawing on data from a large representative sample of the population

- Examples from the social sciences
  - Gallup World Poll
  - World Values Survey / European Values Study
  - European Social Survey
  - Eurobarometer
  - Latinobarometro
  - Afrobarometer
  - Asia Barometer, Asian Barometer

# Population and sample

- Population
  - complete set of all items (e.g. individuals) of research interest

- Sample
  - set of items selected from the population



# Sampling techniques

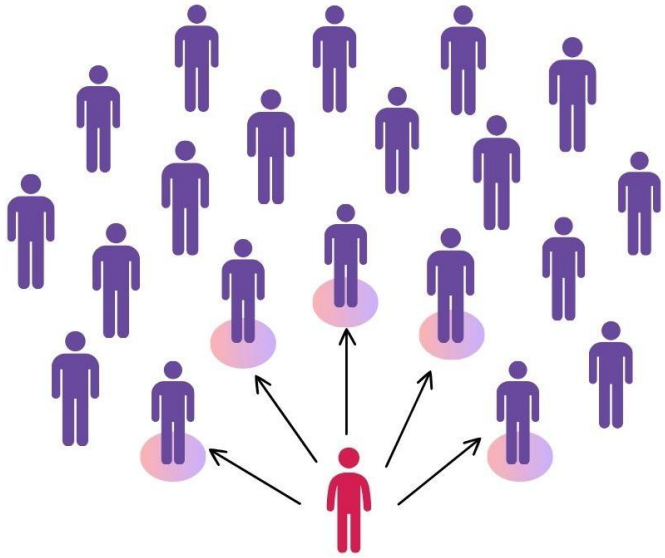
| | Non-probability sampling | Probability sampling |
|---|---|---|
| Inclusion probability | unknown | known, non-zero |
| Pros | lower costs<br>time efficient | higher costs<br>time consuming |
| Cons | selection bias<br>no generalizability | no selection bias<br>generalizability |
| Common types | Convenience<br>Snowball<br>Quota<br>Purposive | Simple random<br>Stratified random<br>Systematic<br>Cluster |

# Non-probability sampling: Convenience sample

- Select participants who are easy to reach
- Example:
  - Psychology professor recruits subjects for a lab test among class participants

- Pros:
  - high proximity of participants
  - quick
  - cheap

- Cons:
  - selection bias
  - no generalizability

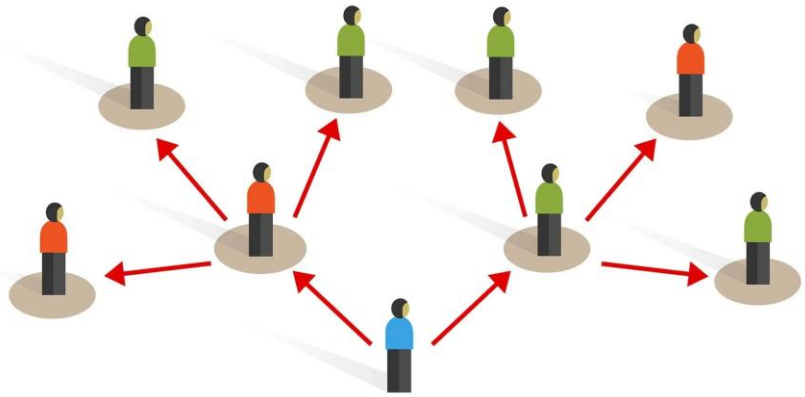


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# Non-probability probability: Snowball sample

- Researcher recruits a number of initial participants (initial seed) who recruit further participants from their social networks (referral); newly recruited participants recruit further ones (chain expansion) )until desired sample size has been reached
- Example: Recruiting subjects for a study on illegal drug use or rare medical conditions

- Pros:
  - Access to hidden groups
  - Built-in trust
  - Cheap and quick
- Cons:
  - selection bias
  - no generalizability
  - Exclusion of isolates



https://www.simplypsychology.org/wp-content/uploads/snowball-sampling.jpg

# Non-probability sampling: Quota sample

- Divide population into mutually exclusive subgroups based on key characteristics known in advance (e.g. from census), set fixed sizes (quotas) for each group (proportional or non-proportional to its population share), select (typically via availability or convenience) until each quota is full
- Example: Budget-constrained social science study

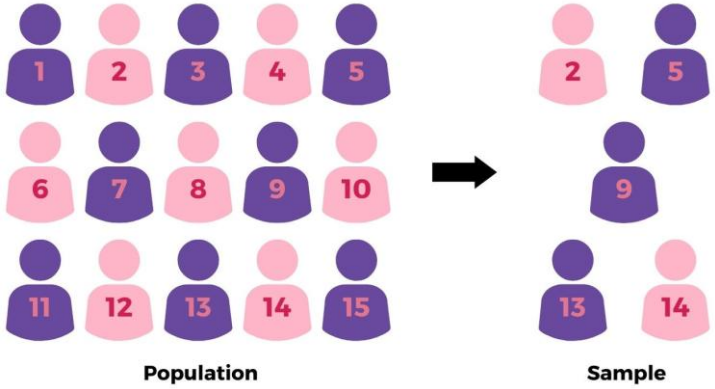
- Pros:
  - Cheap and quick
  - Ensures specific subgroups
- Cons:
  - Selection bias
  - Limited generalizability

| Group | Population share | Quota (*N*) |
|---|---|---|
| Male | 49.38 % | 98 |
| Female | 50.62 % | 102 |
| Total | 100.00 % | 200 |

# Probability sampling: Simple random sample

- Every member of the population has an equal chance of being selected
- From a complete, numbered list of every member of the population, select the needed number of cases at random (e.g. random number generator, lottery)

- Pros
  - No selection bias
  - Generalizable
- Cons
  - Costly, time-consuming
  - Needs complete list of population

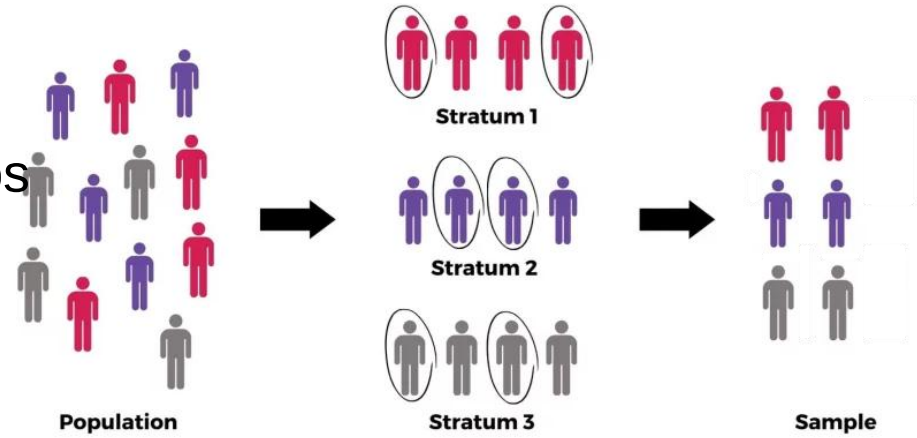



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# Probability sampling: Stratified random sample

- Divide the population into strata – mutually exclusive homogeneous subgroups, based on shared demographics (e.g. gender, age, region)
- Randomly select members from each strata
  - Proportionate stratification: stratum size matches population share
  - Disproportionate stratification: oversampling smaller groups

- Pros:
  - Ensures major subgroups
  - Ensures minority subgroups

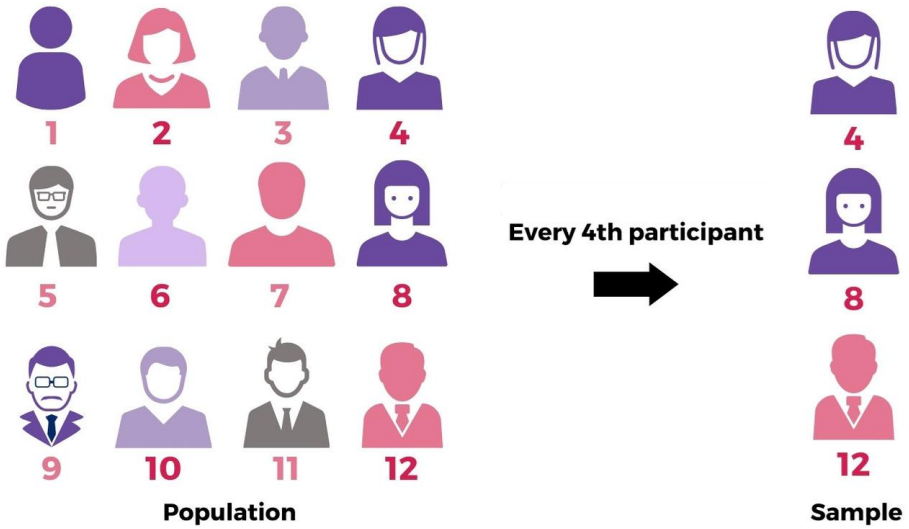



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# Probability sampling: Systematic sample

- Select members from the population at a regular, fixed interval given a random starting point

- Pros:
  - Simple and fast
- Cons:
  - Complete list of population
  - Ordered sampling frame




https://tgmresearch.com/images/library/blog-2023/systematic-sampling/systematic-random-sampling.jpg

# Representative sample

Non-biased statistical description and inference are only possible with representative samples!

- Characteristics of the cases in the sample are approximately the same as those of the population
- Representativity must be defined
  - time: e.g. 2026
  - space: e.g. Germany
  - matter: e.g. biological sex, age, education, migration background

- Can be best achieved via probability sampling

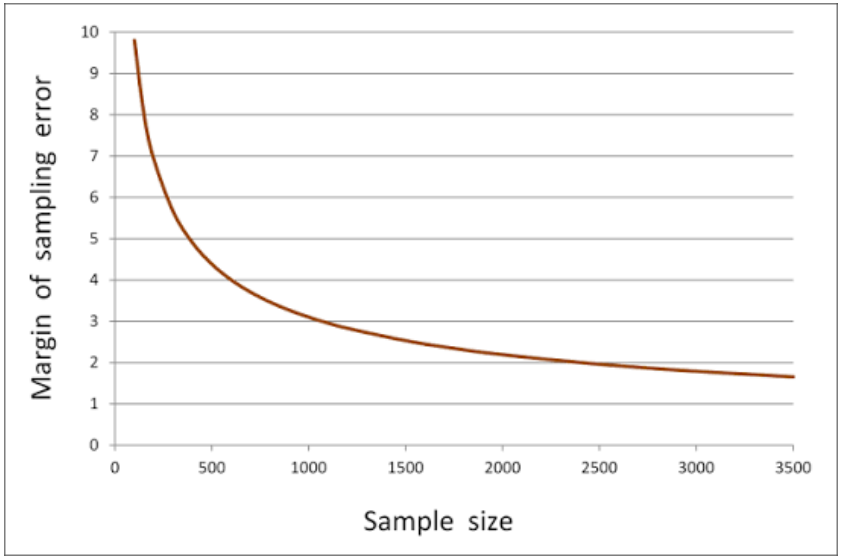
- Weights can be applied after sampling
  - Design weight: adjust probability of selection of individual cases
  - Calibration weight: adjust sample to population

# How large is large enough?

- Central Limit Theorem
  - $N > 30$ for an (approximately) normal distribution
- Sample size not that important for representativity but very important for precision and power

- Standard error of the…
  - mean
  - proportion

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \qquad SE_p = \sqrt{\frac{p(1-p)}{n}}$$




The higher the sample size, the smaller the standard error, i.e. the higher the precision with which the sample mean / proportion estimate the 'true' population mean / proportion.



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